

Section 3.5

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$$3. \frac{dy}{dx} = x^2(-\sin x) + 2x \cdot \cos x = -x^2 \sin x + 2x \cos x$$

$$12. \frac{dy}{dx} = \frac{(1+\sin x) \frac{d}{dx}(\cos x) - (\cos x) \frac{d}{dx}(1+\sin x)}{(1+\sin x)^2} = \frac{(1+\sin x)(-\sin x) - (\cos x) \cos x}{(1+\sin x)^2} = \frac{-\sin x - 1}{(1+\sin x)^2} = -\frac{1}{1+\sin x}$$

$$15. \frac{dy}{dx} = \frac{d}{dx}(\sec^2 x - \tan^2 x) = \frac{d}{dx} 1 = 0$$

$$17. \frac{d^2 f}{dx^2} = x^3 \frac{d}{dx}(\sin x \cos x) + 3x^2 \sin x \cos x$$

$$= x^3(-\sin^2 x + \cos^2 x) + 3x^2 \sin x \cos x$$

$$26. \frac{df}{d\theta} = (1+\sec \theta)(\cos \theta) + (\sec \theta \tan \theta)(\sin \theta)$$

$$= \cos \theta + 1 + \tan^2 \theta = \cos \theta + \sec^2 \theta$$

$$32. \frac{dp}{dq} = \frac{(q \sec q)(3 + \sec^2 q) - (3q + \tan q)(q \sec q \tan q + \sec q)}{(q \sec q)^2}$$

$$34. a) y^{(4)} = \frac{d^3}{dx^3}(-2\cos x) = \frac{d^2}{dx^2}(2\sin x) = \frac{d}{dx}(2\cos x) = -2\sin x$$

$$b) y^{(4)} = \frac{d^3}{dx^3}(-9\sin x) = \frac{d^2}{dx^2}(-9\cos x) = \frac{d}{dx}(9\sin x) = 9\cos x$$

$$45. a) m = \frac{dy}{dx} \Big|_{x=\pi/2} = -\csc^2 x + 2\csc x \cot x \Big|_{x=\pi/2} = -\csc^2 \frac{\pi}{2} + 2\csc \frac{\pi}{2} \cot \frac{\pi}{2} = -1$$

The equation will be $y - 2 = -1(x - \frac{\pi}{2})$ or $y = -x + (\frac{\pi}{2} + 2)$

$$b) \text{ At point } Q, \frac{dy}{dx} = 0.$$

$$\frac{dy}{dx} = -\csc^2 x + 2\csc x \cot x$$

$$0 = \csc x(-\csc x + 2\cot x)$$

$\therefore$ Since $\csc x \neq 0$, we get that $\csc x = 2\cot x$

$$1 = 2\cos x \rightarrow x = \frac{\pi}{3}$$

The horizontal tangent to the curve at Q will be $y = 4 + \cot \frac{\pi}{3} - 2\csc \frac{\pi}{3} = 4 + \frac{1}{\sqrt{3}} - \frac{4}{\sqrt{3}} = 4 - \sqrt{3}$

$$50. \lim_{\theta \rightarrow \pi/4} \frac{\tan \theta - 1}{\theta - \frac{\pi}{4}} = \frac{d}{d\theta} \tan \theta \Big|_{\theta = \frac{\pi}{4}} = \sec^2 \frac{\pi}{4} = (\sqrt{2})^2 = 2$$

$$51. \lim_{x \rightarrow 0} \sec \left[\cos x + \pi \tan \left(\frac{\pi}{4 \sec x} \right) - 1 \right] = \sec \left[1 + \pi \tan \left(\frac{\pi}{4} \right) - 1 \right] = \sec \pi = -1$$

$$56. v(t) = \frac{ds}{dt} = \cos t - \sin t \rightarrow v\left(\frac{\pi}{4}\right) = \cos \frac{\pi}{4} - \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0 \text{ m/s}$$

$$a(t) = \frac{dv}{dt} = -\sin t - \cos t \rightarrow a\left(\frac{\pi}{4}\right) = -\sin \frac{\pi}{4} - \cos \frac{\pi}{4} = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2} \text{ m/s}^2$$

$$j(t) = \frac{da}{dt} = -\cos t + \sin t \rightarrow j\left(\frac{\pi}{4}\right) = -\cos \frac{\pi}{4} + \sin \frac{\pi}{4} = -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = 0 \text{ m/s}^3$$

$$58. \text{ Continuous: } \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x)$$

$$b = \cos 0 = 1$$

$$\text{Check differentiable: } \lim_{x \rightarrow 0^-} \frac{g(0) - g(x)}{-x} = \lim_{x \rightarrow 0^-} \frac{1 - (x+1)}{-x} = 1$$

(for $b=1$)

$$\lim_{x \rightarrow 0^+} \frac{g(x) - g(0)}{x} = \frac{d}{dx}(\cos x) \Big|_{x=0} = -\sin(0) = 0 \rightarrow \text{Not differentiable at } x=0$$

$$59. \text{ Since } \frac{d^4}{dx^4}(\cos x) = \cos x, \text{ we get } \frac{d^{999}}{dx^{999}}(\cos x) = \frac{d^3}{dx^3}(\cos x) = \sin x$$

Section 3.6

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$$5. \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du}(\sqrt{u}) \cdot \frac{d}{dx}(\sin x) = \frac{1}{2\sqrt{u}} \cdot \cos x = \frac{\cos x}{2\sqrt{\sin x}}$$

$$7. \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du}(\tan u) \cdot \frac{d}{dx}(\pi x^2) = \sec^2 u \cdot 2\pi x = 2\pi x \cdot \sec^2(\pi x^2)$$

$$10. y = u^9, u = 4-3x$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 9u^8 \cdot (-3) = -27u^8 = -27(4-3x)^8$$

$$15. y = \sec u, u = \tan x$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = -\sec u \tan u \cdot \sec^2 x = -\sec(\tan x) \tan(\tan x) \sec^2 x$$

$$29. \frac{dy}{dx} = (4x+3)^4 \cdot \frac{d}{dx}(x+1)^{-3} + (x+1)^{-3} \cdot \frac{d}{dx}(4x+3)^4$$

$$= (4x+3)^4 \cdot (-3)(x+1)^{-4}(1) + (x+1)^{-3} \cdot 4(4x+3)^3 \cdot 4$$

$$= -3 \cdot (4x+3)^4 (x+1)^{-4} + 16(4x+3)^3 (x+1)^{-3}$$

$$34. \frac{dy}{dx} = \frac{(x+7)^4 (\sec^2 3x \cdot 3) - \tan 3x \cdot (4(x+7)^3(1))}{(x+7)^8} = \frac{3(x+7)^4 (\sec^2 3x) - 4(\tan 3x)(x+7)^3}{(x+7)^8}$$

$$41. \frac{dy}{dt} = 2 \sin(\pi t - 2) \cdot \cos(\pi t - 2) \cdot \pi = 2\pi \sin(\pi t - 2) \cos(\pi t - 2)$$

$$54. \frac{dy}{dt} = 4 \cos(\sqrt{1+t}) \left(\frac{1}{2\sqrt{1+t}} \right) \left(\frac{1}{2t} \right)$$

$$65. (f \circ g)'(1) = \frac{df}{du} \cdot \frac{du}{dx} \Big|_{x=1} = (5u^4) \left(\frac{1}{2\sqrt{x}} \right) \Big|_{x=1} = (5(\sqrt{x})^4) \left(\frac{1}{2\sqrt{x}} \right) \Big|_{x=1} = 5 \cdot (\sqrt{1})^4 \cdot \frac{1}{2\sqrt{1}} = \frac{5}{2}$$

$$70. (f \circ g)'(-1) = \frac{df}{du} \cdot \frac{du}{dx} \Big|_{x=-1} = 2 \left(\frac{u-1}{u+1} \right) \left(\frac{2}{(u+1)^2} \right) \left(-\frac{2}{x^3} \right) \Big|_{x=-1} = 2 \left(\frac{-1}{1} \right) \left(\frac{2}{1^2} \right) \left(-\frac{2}{(-1)^3} \right) = 8$$

$$73. a) \frac{d}{dx}(2f(x)) \Big|_{x=2} = 2f'(2) = 2\left(\frac{1}{2}\right) = \frac{1}{2}$$

$$b) \frac{d}{dx}(f(x)+g(x)) \Big|_{x=3} = f'(3) + g'(3) = 2\pi + 5$$

$$c) \frac{d}{dx}(f(x) \cdot g(x)) \Big|_{x=3} = f(3)g'(3) + g(3)f'(3) = 3 \cdot 5 + (-4)(2\pi) = 15 - 8\pi$$

$$d) \frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) \Big|_{x=2} = \frac{g(2)f'(2) - f(2)g'(2)}{g(2)^2} = \frac{2\left(\frac{1}{2}\right) - 8(-3)}{2^2} = \frac{37}{6}$$

$$e) \frac{d}{dx}(f(g(x))) \Big|_{x=2} = f'(g(2))g'(2) = f'(2)g'(2) = \frac{1}{3} \cdot (-3) = -1$$

$$f) \frac{d}{dx}(\sqrt{f(x)}) \Big|_{x=2} = \frac{1}{2\sqrt{f(2)}} \cdot f'(2) = \frac{1}{2\sqrt{8}} \cdot \frac{1}{3} = \frac{1}{6\sqrt{2}}$$

$$g) \frac{d}{dx} \left(\frac{1}{(g(x))^2} \right) \Big|_{x=3} = -\frac{2}{(g(3))^3} \cdot g'(3) = -\frac{2}{(-9)^3} \cdot 5 = \frac{5}{32}$$

$$h) \frac{d}{dx} \left(\frac{f(x)}{\sqrt{f(x)^2 + g(x)^2}} \right) \Big|_{x=2} = \frac{\frac{1}{\sqrt{f(x)^2 + g(x)^2}} \cdot f'(2) - f(2) \cdot \frac{1}{2\sqrt{f(x)^2 + g(x)^2}} \cdot 2g(2)g'(2)}{\sqrt{f(x)^2 + g(x)^2}^2} = \frac{1}{2\sqrt{8^2 + 2^2}} (2 \cdot 8 \cdot \frac{1}{3} + 2 \cdot 2 \cdot (-3)) = \frac{1}{4\sqrt{17}} \left(-\frac{20}{3} \right) = -\frac{5}{3\sqrt{17}}$$

$$74. a) \frac{d}{dx}(5f(x) - g(x)) \Big|_{x=1} = 5f'(1) - g'(1) = 5\left(\frac{1}{2}\right) - \left(-\frac{8}{3}\right) = 1$$

$$b) \frac{d}{dx}(f(x)g^2(x)) \Big|_{x=0} = f(0) \cdot 2g(0)g'(0) + f'(0)g^2(0) = 1 \cdot 3(1^2) \left(\frac{1}{3}\right) + 5(1^3) = 6$$

$$c) \frac{d}{dx} \left(\frac{f(x)}{g(x)+1} \right) \Big|_{x=1} = \frac{(g(1)+1)f'(1) - f(1)g'(1)}{(g(1)+1)^2} = \frac{(-3)(-\frac{1}{3}) - (3)(-\frac{8}{3})}{(-3)^2} = \frac{9}{9} = 1$$

$$d) \frac{d}{dx}(f(g(x))) \Big|_{x=0} = f'(g(0)) \cdot g'(0) = f'(1) \cdot \frac{1}{3} = -\frac{1}{3} \cdot \frac{1}{3} = -\frac{1}{9}$$

$$e) \frac{d}{dx}(g(f(x))) \Big|_{x=0} = g'(f(0)) \cdot f'(0) = g'(1) \cdot 5 = -\frac{8}{3} \cdot 5 = -\frac{40}{3}$$

$$f) \frac{d}{dx}(x^4 + f(x))^{-2} \Big|_{x=1} = -2(x^4 + f(x))^{-3} \cdot (4x^3 + f'(x)) \Big|_{x=1} = -2(1+3)^{-3} (11 - \frac{1}{3}) = -\frac{2}{64} \cdot \frac{32}{3} = -\frac{1}{3}$$

$$g) \frac{d}{dx}(f(x+g(x))) \Big|_{x=0} = f'(x+g(x)) \cdot (1+g'(x)) = f'(1) \cdot (1 + \frac{1}{3}) = -\frac{1}{3} \cdot \frac{4}{3} = -\frac{4}{9}$$

Section 3.6 (Continue)

82. a) $y = \sin 2x$: $m = \frac{dy}{dx} = 2\cos 2x \rightarrow \text{Slope at } (0,0) = 2\cos 0 = 2 \rightarrow \text{Equation of the tangent line: } y = 2x$
 $y = -\sin \frac{x}{2}$: $m = \frac{dy}{dx} = -\frac{1}{2}\cos \frac{x}{2} \rightarrow \text{Slope at } (0,0) = -\frac{1}{2}\cos 0 = -\frac{1}{2} \rightarrow \text{Equation of the tangent line: } y = -\frac{x}{2}$

Both tangents' slopes multiplied to be -1 , which means they are perpendicular to each other.

b) $y = \sin mx$, $y' = m\cos mx \rightarrow y'(0) = m\cos 0 = m \rightarrow \text{tangent at the origin is } y = mx$
 $y = -\sin \frac{x}{m}$, $y' = -\frac{1}{m}\cos \frac{x}{m} \rightarrow y'(0) = -\frac{1}{m}\cos 0 = -\frac{1}{m} \rightarrow \text{tangent at the origin is } y = -\frac{1}{m}x$
 $\therefore$ Both tangents' slope multiplied to be -1 , which means they are perpendicular to each other.

c) $y = \sin mx \rightarrow y' = m\cos mx \rightarrow -|m| \leq m\cos mx \leq |m| \therefore \text{largest value of the slope} = |m|$
 $y = -\sin \frac{x}{m} \rightarrow y' = -\frac{1}{m}\cos \frac{x}{m} \rightarrow -\frac{1}{|m|} \leq -\frac{1}{m}\cos \frac{x}{m} \leq \frac{1}{|m|} \therefore \text{largest value of the slope} = \frac{1}{|m|}$

d) $y = \sin mx$ will complete m periods on the interval $[0, 2\pi]$

As $y = \sin mx \rightarrow y' = m\cos mx \therefore$ at origin, $y' = m\cos 0 = m$

Hence, the slope of the curve $y = \sin mx$ at the origin is equal to the number of periods $y = \sin mx$ completes on $[0, 2\pi]$

84. a) $y' = 20\cos\left(\frac{2\pi}{365}(x-101)\right) \cdot \frac{2\pi}{365}$

y' will be greatest when $\cos\left(\frac{2\pi}{365}(x-101)\right) = 1$ or $x = 101 \rightarrow \text{Temperature increases fastest on day 101.}$

b) $y'(101) = 20 \cdot 1 \cdot \frac{2\pi}{365} = \frac{40\pi}{365} \approx 0.344^\circ\text{C/day}$

Section 3.7

1. $x^2y + xy^2 = 6$

$$x^2 \frac{dy}{dx} + 2xy + y^2 + 2xy \frac{dy}{dx} = 0$$

$$(x^2 + 2xy) \frac{dy}{dx} + (2xy + y^2) = 0$$

$$\frac{dy}{dx} = \frac{-2xy - y^2}{x^2 + 2xy}$$

12. $x^4 + \sin y = x^3y^2$

$$4x^3 + \cos y \frac{dy}{dx} = 3x^2y^2 + 2x^3y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{3x^2y^2 - 4x^3}{\cos y - 2x^3y}$$

19. $x^2 + y^2 = 1$

$$2x + 2y \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

$$2 + 2y \frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 0$$

$$2y \frac{d^2y}{dx^2} = -2 - 2\left(\frac{dy}{dx}\right)^2 = -2 - 2\left(-\frac{x}{y}\right)^2 = -2 - \frac{2x^2}{y^2}$$

$$\frac{d^2y}{dx^2} = \frac{-2 - \frac{2x^2}{y^2}}{2y}$$

Section 3.7 (Continue)

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29. $x^2 + xy - y^2 = 1$

Point $(2,3) \rightarrow 2^2 + 2(3) - 3^2 = 4 + 6 - 9 = 1 \therefore$ The point $(2,3)$ is on the curve.

a) $\frac{dy}{dx}(x^2 + xy - y^2) = \frac{dy}{dx}(1)$

$$2x + x \frac{dy}{dx} + y - 2y \frac{dy}{dx} = 0$$

$$(x-2y) \frac{dy}{dx} = -y-2x$$

$$\frac{dy}{dx} = \frac{-y-2x}{x-2y}$$

Slope of tangent line at $(2,3) = \frac{dy}{dx}|_{x=2, y=3} = \frac{-3-2(2)}{2-2(3)} = \frac{7}{4}$

$\therefore$ Tangent line: $y-3 = \frac{7}{4}(x-2) \rightarrow y = \frac{7}{4}x - \frac{1}{2}$

b) Line that is normal to the curve: slope = $-\frac{1}{\frac{7}{4}} = -\frac{4}{7} \rightarrow$ equation $y-3 = -\frac{4}{7}(x-2)$
 $y = -\frac{4}{7}x + \frac{29}{7}$

41. $y^4 = y^2 - x^2$

$$4y^3 \frac{dy}{dx} = 2y \frac{dy}{dx} - 2x$$

$$2x = (2y - 4y^3) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{2y - 4y^3}{2x} = \frac{y - 2y^3}{x}$$

Slope at $(\frac{\sqrt{3}}{4}, \frac{1}{2}) = \frac{\frac{1}{2} - 2(\frac{\sqrt{3}}{4})^3}{\frac{\sqrt{3}}{4}} = \frac{-\frac{\sqrt{3}}{4}}{\frac{\sqrt{3}}{4}} = -1$

Slope at $(\frac{\sqrt{3}}{4}, \frac{1}{2}) = \frac{\frac{1}{2} - 2(\frac{1}{2})^3}{\frac{\sqrt{3}}{4}} = \frac{1}{\sqrt{3}}$

Section 3.8

7.

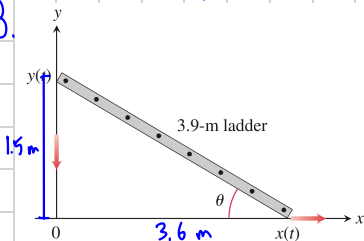
$$x^2 + y^2 = 9y$$

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$9(3)(-2) + 2(-4)(\frac{dy}{dt}) = 0$$

$$\frac{dy}{dt} = -\frac{12}{8} = -\frac{3}{2}$$

93.



$$x^2 + y^2 = 3.9^2$$

Know: $\frac{dx}{dt} = 1.5$ m/s when $x = 3.6$ m

Want: $\frac{dy}{dt}$, $\frac{d}{dt}(\frac{1}{2}xy)$, $\frac{d\theta}{dt}$

a) $\frac{d}{dt}(x^2 + y^2) = \frac{d}{dt}(3.9^2)$

$$\frac{dx}{dt} \cdot \frac{dx}{dt} + \frac{dy}{dt} \cdot \frac{dy}{dt} = 0$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = \frac{-x \cdot \frac{dx}{dt}}{y} = \frac{-3.6 \cdot 1.5}{\sqrt{3.9^2 - 3.6^2}} = -3.6 \text{ m/s} \rightarrow 3.6 \text{ m/s down the wall}$$

b) $\frac{d}{dt}(\frac{1}{2}xy) = \frac{1}{2} \frac{d}{dt}(xy)$
 $= \frac{1}{2} (x \frac{dy}{dt} + y \frac{dx}{dt})|_{x=3.6}$
 $= \frac{1}{2} (3.6 \cdot (-3.6) + 1.5(1.5))$
 $= \frac{1}{2} (-10.71) = -5.355 \text{ m}^2/\text{s}$

23. (Continue)

c) From $\cos \theta = \frac{x}{3.9}$
 $\frac{d}{dt}(\cos \theta) = \frac{d}{dt}\left(\frac{x}{3.9}\right)$
 $-\sin \theta \cdot \frac{d\theta}{dt} = \frac{1}{3.9} \cdot \frac{dx}{dt}$
 $\frac{d\theta}{dt} = -\frac{1}{3.9 \sin \theta} \cdot \frac{dx}{dt}$
 $= -\frac{1}{1.5} \times 1.5 = -1 \text{ rad/s}$

27. $V = \frac{1}{3} \pi r^2 h$

$h = \frac{3}{8} r$

$\frac{dV}{dt} = 10 \text{ m}^3/\text{min}$

When $h=4$, we get $r = 4 \times \frac{8}{3} = \frac{32}{3} \text{ m}$

From $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi r^2 \left(\frac{3}{8} r\right) = \frac{1}{8} \pi r^3$

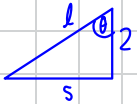
$\frac{dV}{dt} = \frac{1}{8} \pi \cdot 3r^2 \frac{dr}{dt}$

$10 = \frac{1}{8} \pi \cdot 3 \cdot \left(\frac{32}{3}\right)^2 \cdot \frac{dr}{dt}$

b) $\frac{dr}{dt} = 0.736 \text{ m/min}$

a) $\frac{dh}{dt} = \frac{3}{8} \frac{dr}{dt} = 0.736 \cdot \frac{3}{8} = 0.276 \text{ m/min}$

32.



Know: $l^2 - s^2 = 4$, $\frac{dl}{dt} = -0.5$

a) Finding $\left.\frac{ds}{dt}\right|_{l=3} \rightarrow$ at $l=3$, $s = \sqrt{3^2 - 2^2} = \sqrt{5}$

From $l^2 - s^2 = 4$

$2l \frac{dl}{dt} - 2s \frac{ds}{dt} = 0$

$2 \cdot 3 \cdot (-0.5) - 2(\sqrt{5}) \left(\frac{ds}{dt}\right) = 0$

$\frac{ds}{dt} = -\frac{3}{2\sqrt{5}} = -\frac{3\sqrt{5}}{10} \text{ m/s}$ ($\frac{3\sqrt{5}}{10} \text{ m/s}$ towards the dock)

b) At $l=3$, $\sec \theta = \frac{3}{2}$, $\tan \theta = \frac{\sqrt{5}}{2}$

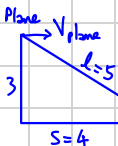
From $\sec \theta = \frac{l}{2}$

$\sec \theta \tan \theta \frac{d\theta}{dt} = \frac{1}{2} \frac{dl}{dt}$

$\frac{3}{2} \cdot \frac{\sqrt{5}}{2} \cdot \frac{d\theta}{dt} = \frac{1}{2} \cdot (-0.5)$

$\frac{d\theta}{dt} = -\frac{\sqrt{5}}{15} \text{ rad/s}$

42.



Know: $l^2 = 3^2 + s^2$, $\frac{dl}{dt} = -160$, $\frac{ds}{dt} = V_{\text{car}} - 120$

$l = \sqrt{9 + s^2}$

$\frac{dl}{dt} = \frac{1}{\sqrt{9 + s^2}} \cdot 2s \cdot \frac{ds}{dt}$

Plug in $l=5$, $s=4 \rightarrow -160 = \frac{4}{\sqrt{9+16}} \cdot \frac{ds}{dt}$

$\frac{ds}{dt} = -200$

$V_{\text{car}} - 120 = 200$

$V_{\text{car}} = 320$ (Towards the car)

$V_{\text{car}} = 80 \text{ km/h}$